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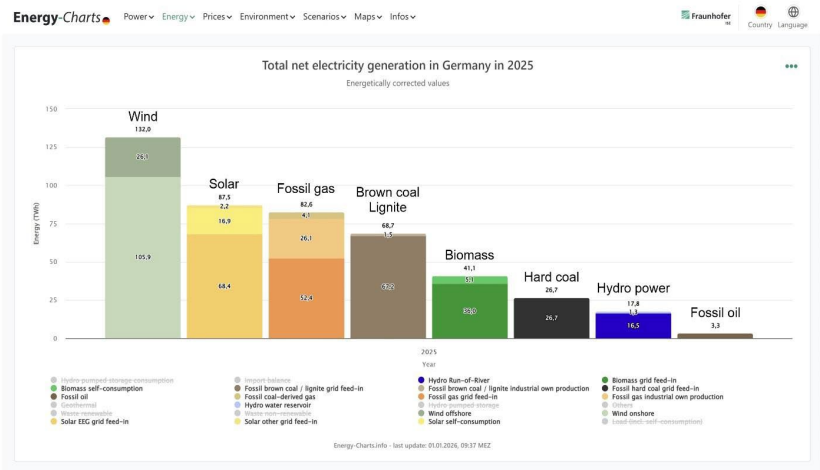
The state of High Temperature Superconductivity in 2025

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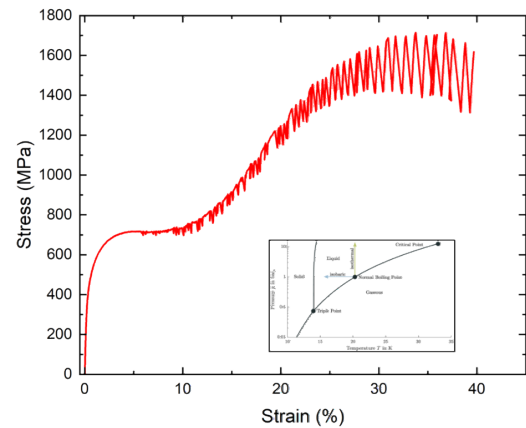
Energy Context

In 2025, Germany reached a significant milestone in its energy transition: for the first time, wind power and photovoltaics became the leading sources of net electricity generation. This development confirms the continuing structural shift of the German power system toward renewable energy sources and underlines the growing need for advanced grid technologies capable of handling high power densities, variability, and increasing electricity demand. In this context, high-temperature superconductivity is gaining strategic relevance as an enabling technology for efficient, compact, and resilient energy infrastructure.



Policy, Regulatory, and Infrastructure

Germany continues to invest in advanced research infrastructures that support the development of next-generation energy technologies, including superconductivity and hydrogen systems. A notable achievement was the successful establishment of the first material testing environment for liquid hydrogen applications at Karlsruhe Institute of Technology (KIT). Material tests were conducted in liquid hydrogen conditions for a hollow specimen from 20K - 295K at 200 bar pressure, and a fatigue, fracture in situ from 20K - 295K at 1 bar pressure. These facilities enable fatigue and fracture tests under in-situ cryogenic conditions, and represent an important asset for the safe and reliable design of superconducting and hydrogen-compatible components.



R&D Activities

German research institutions and industrial partners remain strongly engaged in applied research on HTS technologies. Within the ROWAMAG project, a superconducting magnet for a magnetic billet heater was successfully tested. The project involves close collaboration between research organisations and industrial partners, including Bültmann, Beck Maschinenfabrik, Theva, and KIT. The newly assembled ROWAMAG magnet, equipped with improved internal current leads, underwent extensive magnet tests with an iron yoke between May and June 2025. Nominal operating conditions were achieved, reaching a current of 505 A, while maintaining stable temperatures below design limits. Following successful validation, the system was transferred to Bültmann in July 2025, marking an important step toward industrial application of HTS-based heating technologies.



Demonstration Projects

Germany continues to play a leading role in large-scale demonstration of superconducting power applications. Within the SUPERLINK project, a major milestone was achieved with the successful pre-qualification test of a 110 kV, 500 MVA superconducting cable system in Munich. The full three-phase system was connected to the high-voltage grid and included all essential components, such as terminations, joints, splitter boxes, as well as cooling, monitoring, and control systems.

The system was operated for more than six months, from October 2024 to June 2025, with controllable load profiles ranging from zero to 500 MVA. Extensive testing included load cycling and sustained operation at high load levels. The results demonstrate the technical maturity, reliability, and grid compatibility of superconducting cable systems for future transmission and distribution networks.



Industrial Players and Activities

The Stella Nova Excellence Center in Hanover opened in 2025, dedicated to research, development, and testing of superconducting cables. This new facility strengthens Germany's position as a European hub for superconducting power technologies and supports innovation for applications ranging from power grids to data centre electrification.

Germany hosts a broad and highly active industrial ecosystem covering the entire HTS value chain.

Key industrial players and technology centres include Adelwitz Technologyzentrum (Torgau), Airbus (Ottobrunn), Bilfinger (Würzburg), Bruker EAS (Hanau), Bültmann (Neuenrade), Evico (Dresden), ERT (Hamburg), FESTO (Esslingen), Messer (Krefeld), Nexans (Hanover), Oswald (Miltgenberg), SHI (Darmstadt), Siemens Energy (Erlangen), Theva (Munich), Trimet (Hamburg), Vision Electric SuperConductors (Kaiserslautern), and Vorbuchner (Kirchweidach).



Success Stories and Lessons Learned

The successful long-term operation of the 110 kV superconducting cable system within the SUPERLINK project represents a major success story for Germany. It demonstrates that HTS technologies are no longer limited to laboratory or pilot-scale environments, but are ready for integration into real-world high-voltage grids. A key lesson was the importance of system-level integration, including reliable cryogenic cooling, monitoring, and control, as well as close cooperation between utilities, manufacturers, and research institutions.

Next Term

In the upcoming term, Germany is expected to further strengthen its focus on system integration, industrialization, and sustainability aspects of HTS technologies. Priorities will include scaling up demonstration projects, improving cost efficiency, enhancing compatibility with hydrogen infrastructures, and supporting the deployment of superconducting solutions in renewable-dominant power systems. Continued collaboration between research organisations, industry, and international partners within the HTS TCP framework will remain central to these efforts.